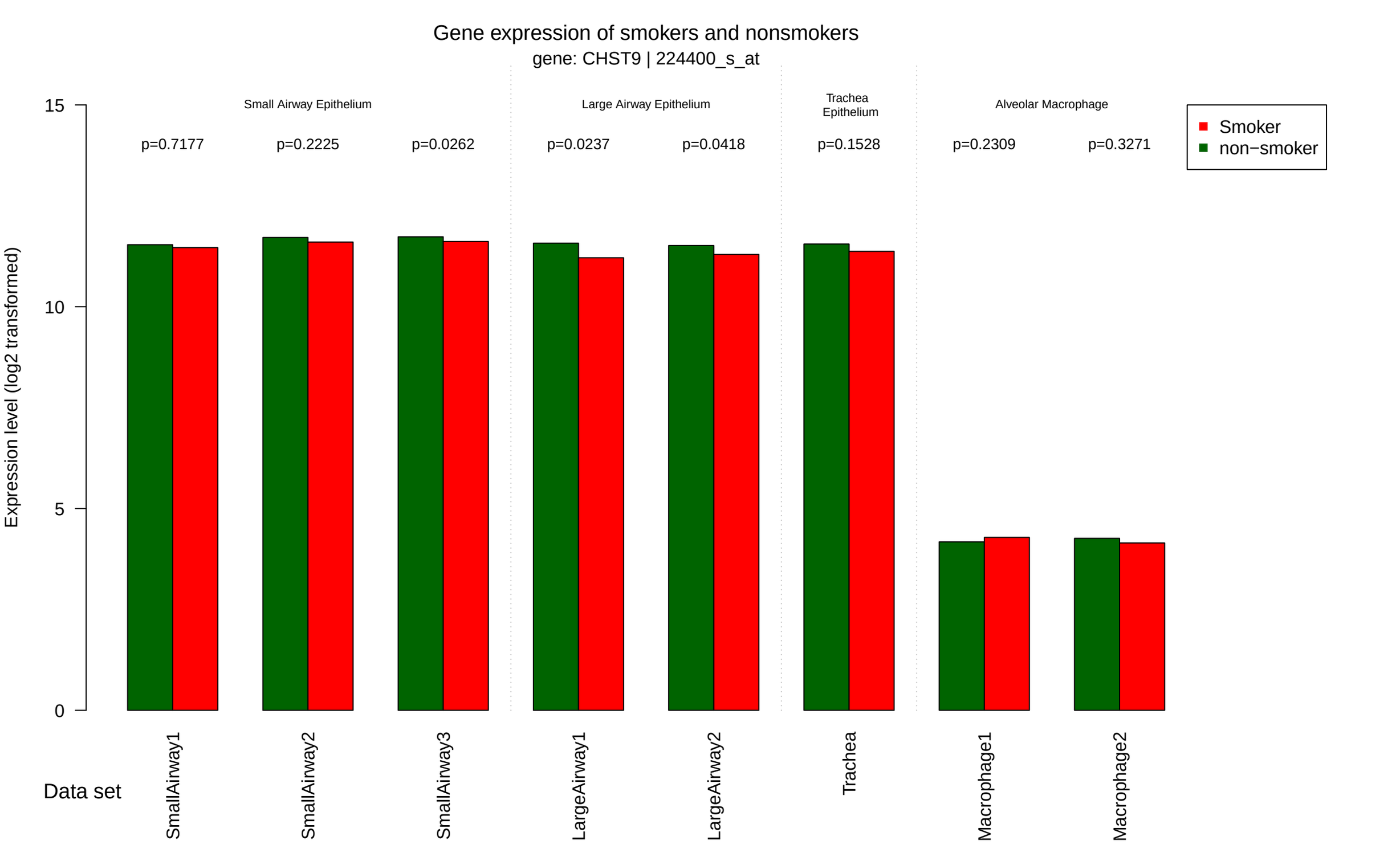
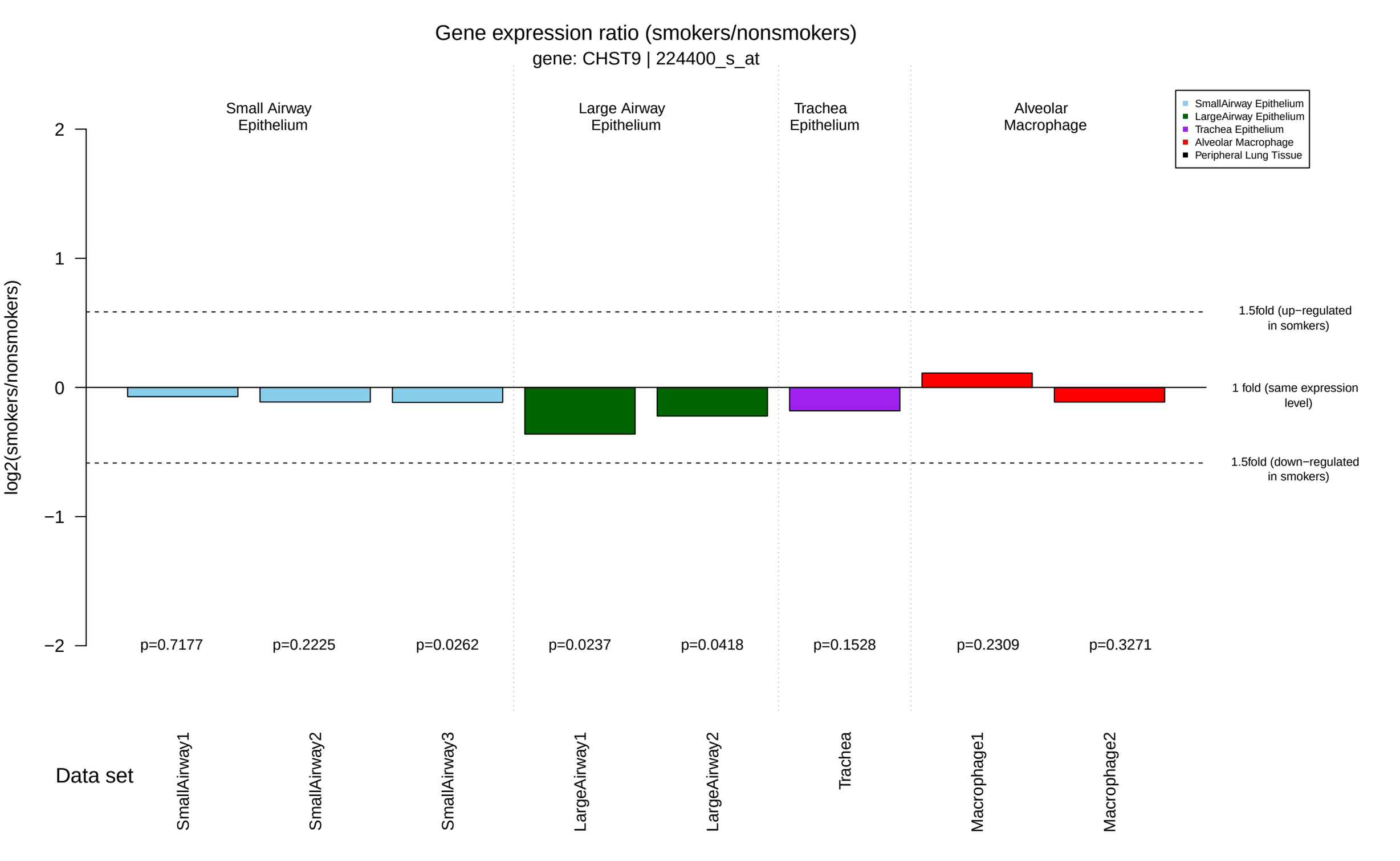
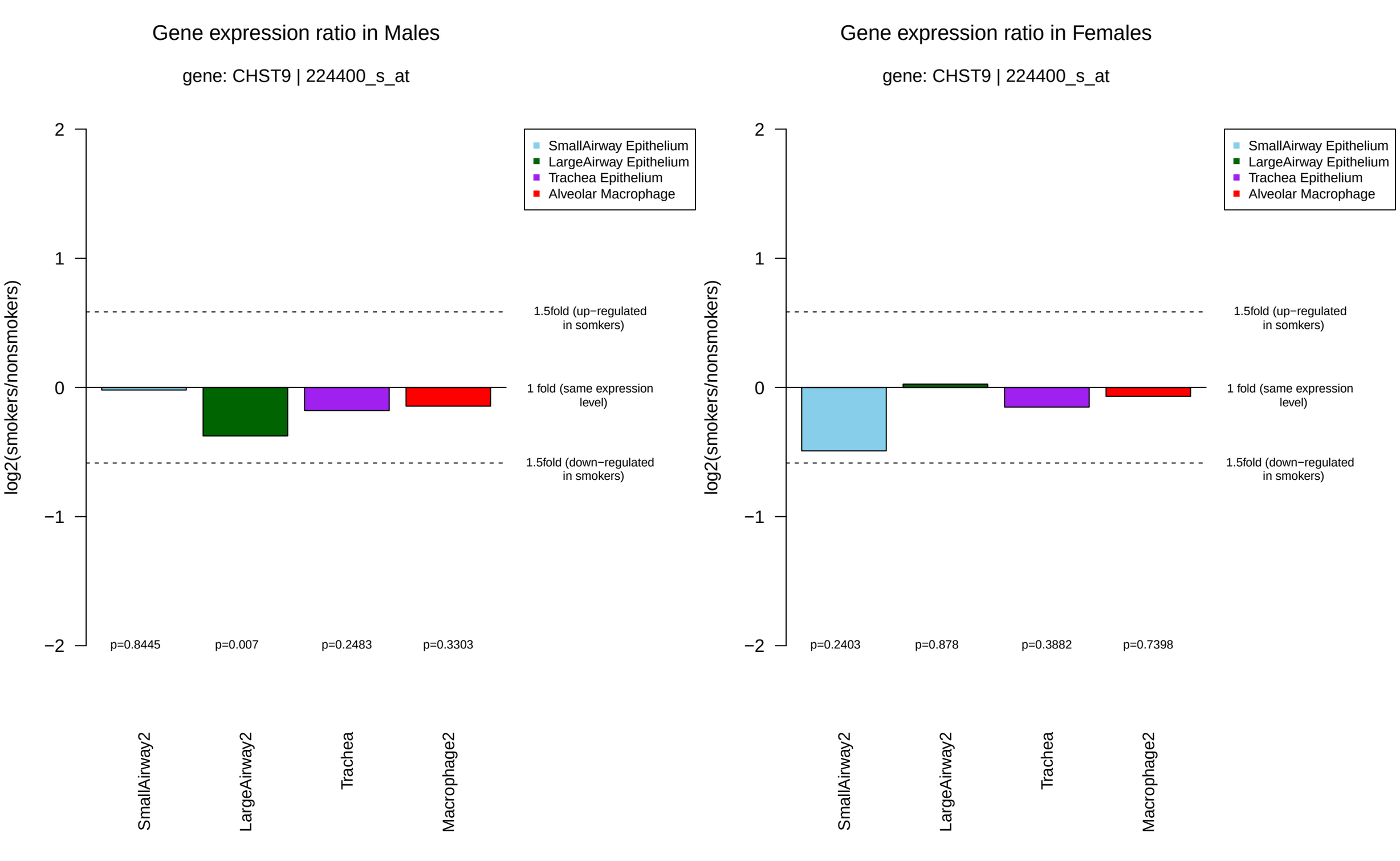
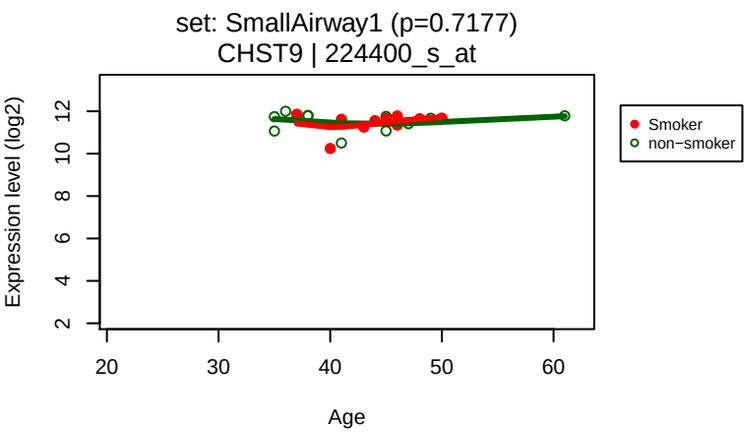
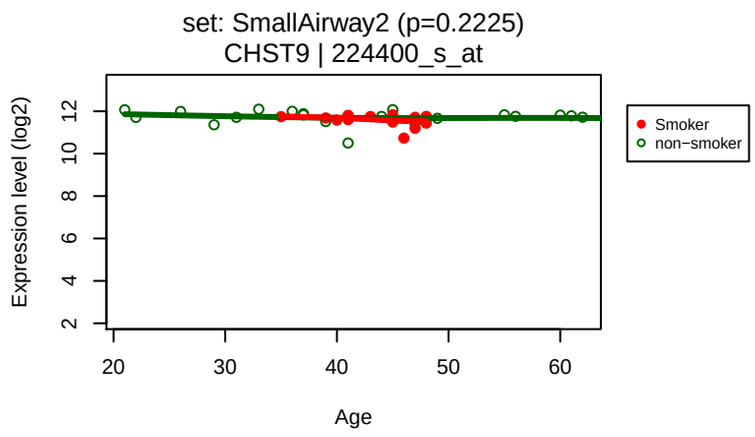
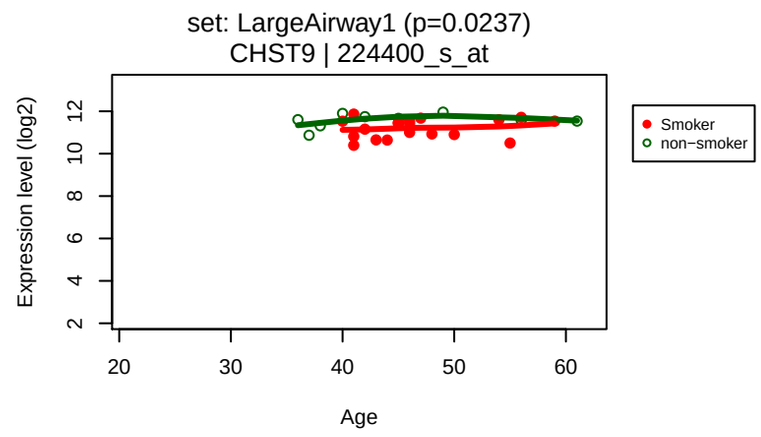

Gene expression of smokers and nonsmokers
gene: CHST9 | 224400_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7177
p=0.2225
p=0.0262
p=0.0237
p=0.0418
p=0.1528
p=0.2309
p=0.3271
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

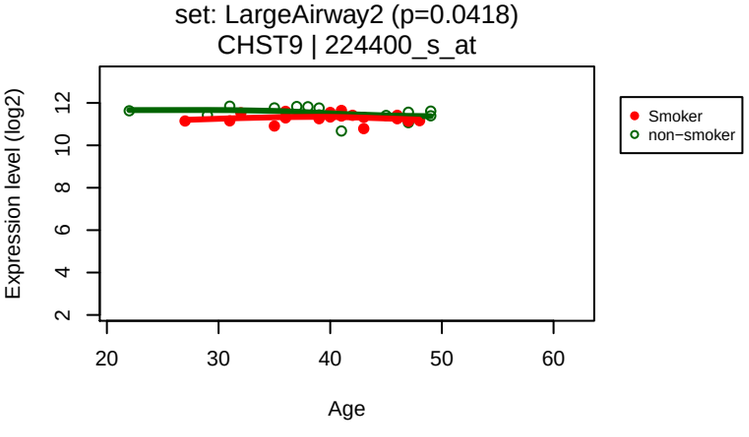

Gene expression ratio (smokers/nonsmokers)
gene: CHST9 | 224400_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7177
p=0.2225
p=0.0262
p=0.0237
p=0.0418
p=0.1528
p=0.2309
p=0.3271
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

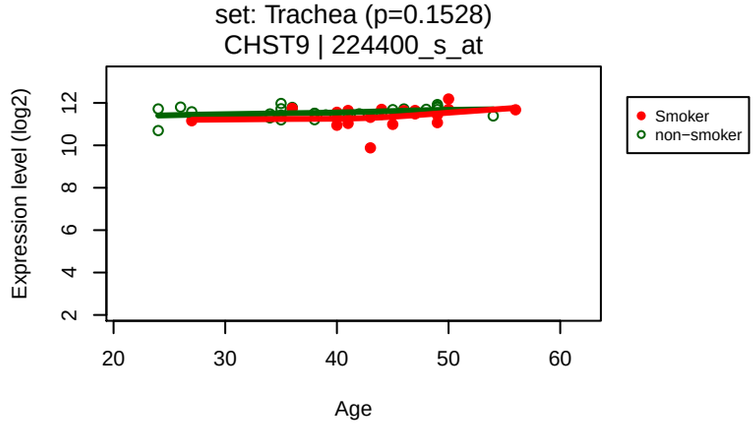

Gene expression ratio in Males
gene: CHST9 | 224400_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8445
p=0.007
p=0.2483
p=0.3303
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHST9 | 224400_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2403
p=0.878
p=0.3882
p=0.7398
SmallAirway2
LargeAirway2
Trachea
Macrophage2

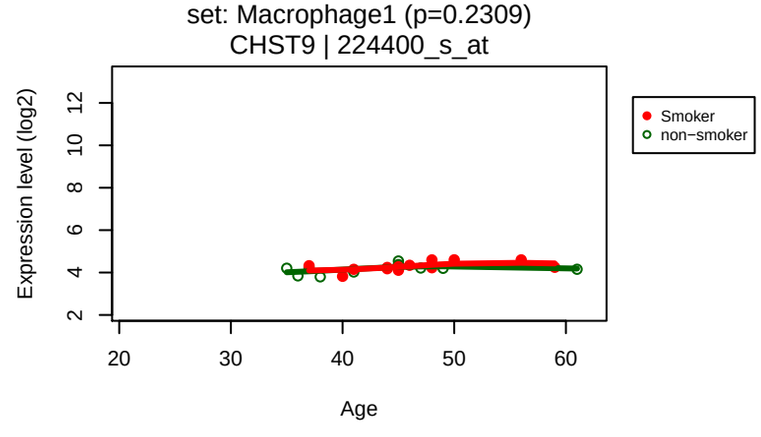

set: SmallAirway1 (p=0.7177)
CHST9 | 224400_s_at
Expression level (log2)
Age
Smoker
non−smoker

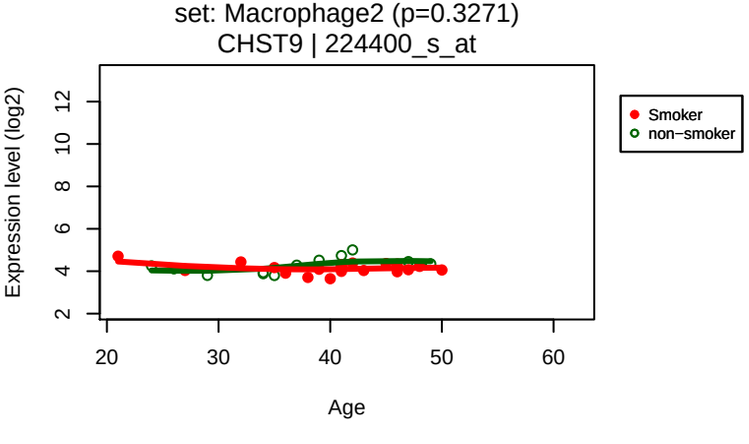

set: SmallAirway2 (p=0.2225)
CHST9 | 224400_s_at
Expression level (log2)
Age
Smoker
non−smoker

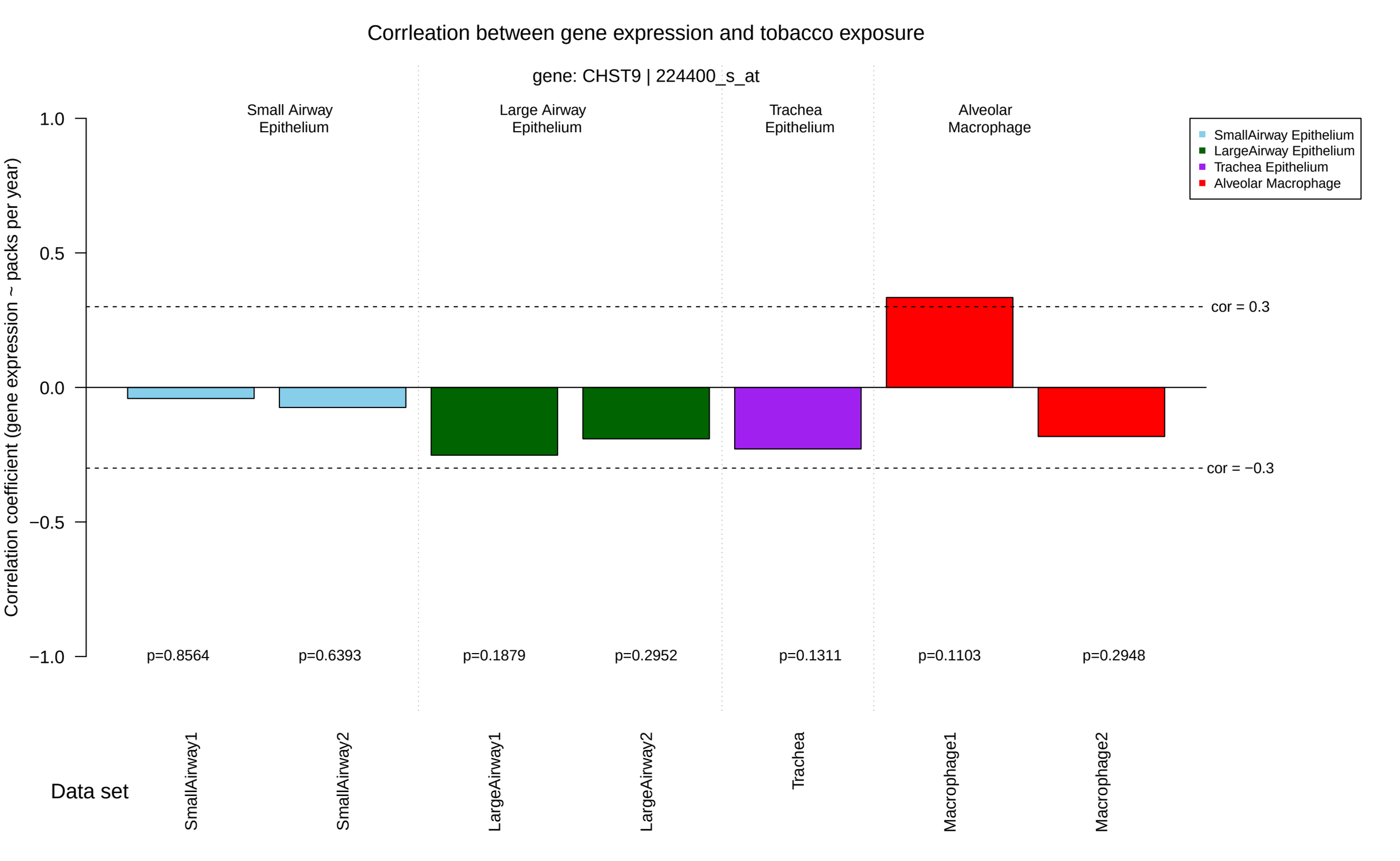

set: LargeAirway1 (p=0.0237)
CHST9 | 224400_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0418)
CHST9 | 224400_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1528)
CHST9 | 224400_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2309)
CHST9 | 224400_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3271)
CHST9 | 224400_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHST9 | 224400_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8564
p=0.6393
p=0.1879
p=0.2952
p=0.1311
p=0.1103
p=0.2948
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2